For my final project, I would like to create a simple circuit solver, similar but more in depth to HW2-Problem 1. This solver would incorporate resistors, inductors, and capacitors, as well as size dynamically based on how many nodes are in the circuit. I chose this project because it directly applies to both of my majors, electrical and computer engineering, and would be extremely useful for modeling analog circuits for any future projects of mine. It is also of a similar structure to a previous homework problem, bridging the worlds of analog and digital via numerical methods, which seemed like the perfect project for the final.

Applied methods are nodal analysis, i.e. Kirchoff's current law to generate a system of equations. From this I can build the matrix that I would need to solve, similarly to HW2, based on the circuit's topology. For the linear system solvers, I will be using gaussian elimination for solving the linear systems, as well as LU decomposition for efficient solving when a circuit has multiple sources.

My project goals are as follows:

- Build a dynamic circuit solver that accepts any user-given topology (utilizing SPICE syntax), computing node voltages & branch currents
- Construct a matrix based on any given circuit's series and parallel configurations
- Support circuits with Resistors, Capacitors, and inductors
- Visualize the circuit topology graphically
- Output all voltage/current results via terminal & graphically, alongside the circuit

The requirements for this project are as follows:

- The solver must correctly compute node voltages for a series and parallel circuit of 3 Resistors with a known voltage source
- The solver must correctly compute node voltages for a series and parallel circuit of 3 capacitors with a known voltage source
- The solver must correctly compute node voltages for a series and parallel circuit of 3 inductors with a known voltage source
- The solver must be able to handle a circuit up to 10 nodes
- The solver must be able to correctly identify and handle any combination of up to 3 series and/or parallel components, properly constructing them in the matrix and giving the correct voltage/current values.